

LionWeb: Standards for Models Interoperability

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Problem

- MDE(/MBSE) brings many advantages
- Some day-to-day problems also arise in the industry:
 - Disparity of tools with no easy path for integration
 - Lack of interoperability and vendor lock-in
 - Obsolete tools
 - Fragmented models



LionWeb

- **Language Interfaces on the Web** — <https://lionweb.io/>
- An open-source project to define
 - standards to make it easy to interchange models between different tools
 - integrate tools in real-time
 - and provide a durable exchange mechanism over time
- Standards are backed by implementations



LionWeb: Repository

Model Repository

Provide an API for:

- storing parts (partitions) of models
- query models
- subscribe to model changes



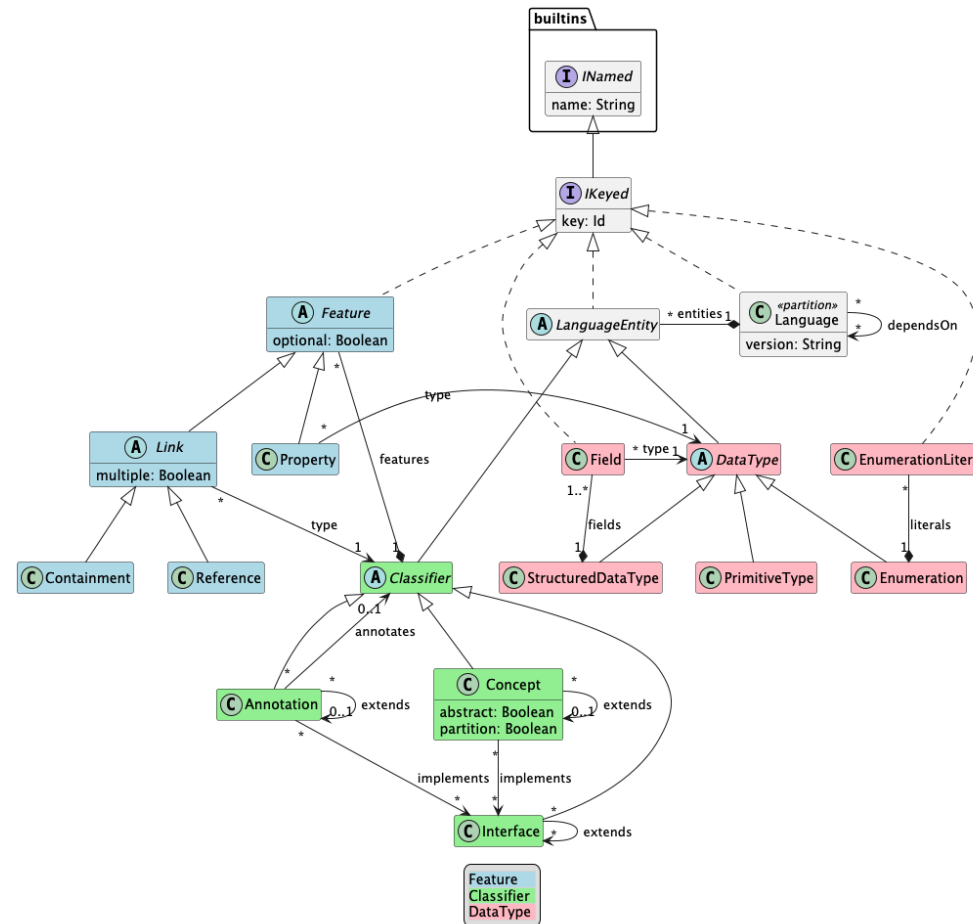
LionWeb: Serialization format

- Serialize models as simple-to-process JSON
 - Works for any modeling language defined through LionCore M3
 - Fault tolerant/resilient



LionWeb: LionCore M3

The meta-metamodel that ensures languages share a common shape.



LionWeb: Bulk Protocol

- For high-efficiency, isolated reads and writes of massive model partitions.



LionWeb: Delta Protocol

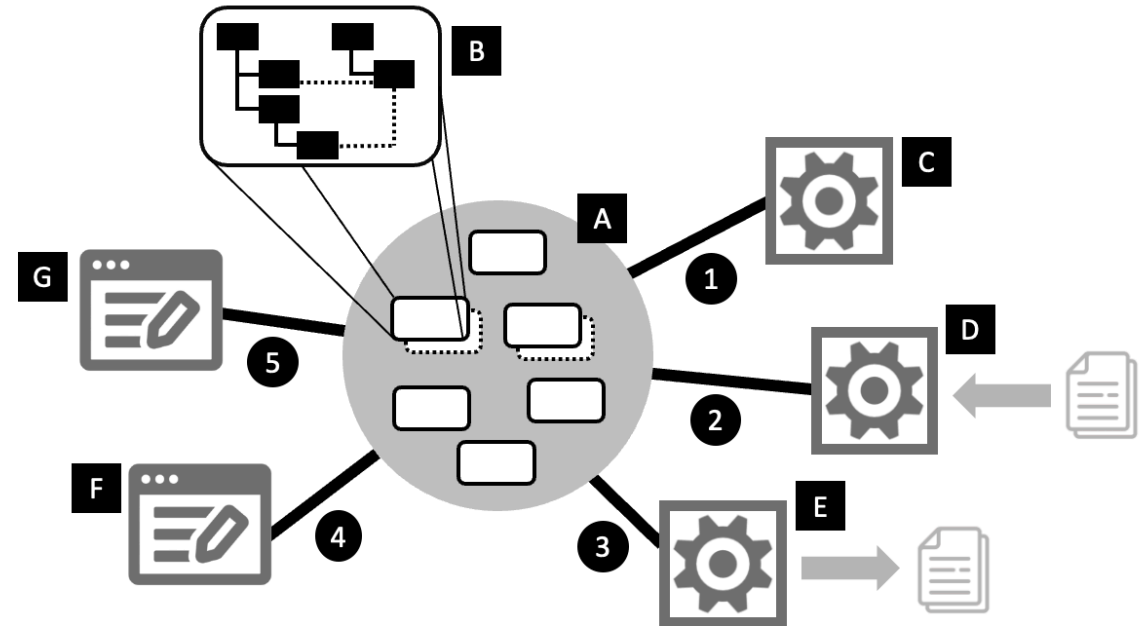
- Continuous, granular updates and real-time, multi-client collaboration.
 - Loosely based on CQRS: queries + commands → events.
 - Can also be used to synchronize a distributed model.



Roadmap

Derived Models

- Enrich existing information suitable for receivers, with traceability
- Formalism: declarative calculations
- Development starting Q4 2026



Demo: space!

Components

- LionWeb server + admin UI — TypeScript
- Model loader — C#
- Model validator — C#
- Simple web client — TypeScript
- Cool web client — TypeScript (Daga)



Enabling technology

Use Cases

1. Integrate modeling tools from different vendors
 - Producers, validation, enrichers, transformations, projections, consumers via Bulk API
2. Integrate real-time modeling tools from different vendors
 - Via Delta Protocol
3. Standardize storage and access for big arbitrary models
 - Reference Repository implementation
4. Migrate models from older versions to new ones in a standardized way
 - Schema Evolution support
5. Ecosystem of tools
 - “LionWeb inside”: Open source and commercial implementations
 - Editors, parsers, model repositories, model-transformation engines, validators, code generators, visual editors, projectional editors, in many stacks and languages. Ensemble your modeling IDE with the suitable COTS to your use case.



Conclusion

- Open, maturing standards for modeling
- Growing ecosystem of tools across platforms:
 - Java, .NET, TypeScript, Kotlin, JetBrains MPS, Python
- Growing support from companies:
 - Strumenta, F1RE, DSL Consultancy, Metadev (Daga), Freon



Call to Action

- Take a look: <https://lionweb.io/>
- Use it: <https://github.com/lionweb-io>
- Collaborate:
 - Slack — join via https://join.slack.com/t/lionweb/shared_invite/zt-2k1k5bsv7-EvC2IABlgSxWrqJNVOZ0HQ
 - Open Office Hours first Friday every month



Q&A

<https://lionweb.io>



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